Indian Institute of Technology Patna

Group 02
Session: 2025 - 2026
Steel Structure Analysis Report



PRATT TRUSS

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OBJECTIVE

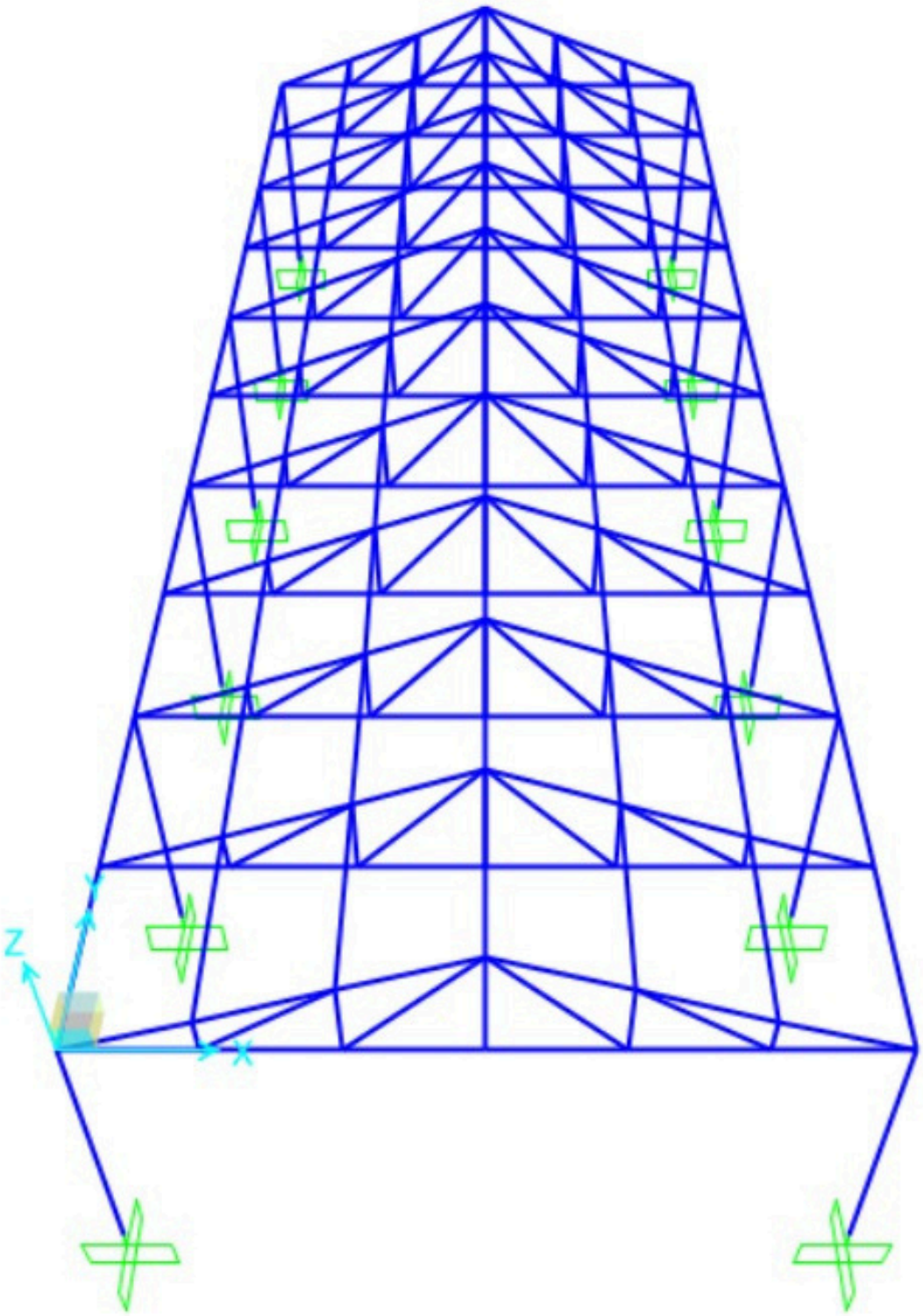
Designing an industrial building roof truss with the following features:

- Proposed location of the industrial building to be designed for 50 years of life – Agartala.
- Dimension of industrial shed – 50 m × 20 m
- Ceiling Height – 10 m (floor to roof truss bottom)
- Spacing of columns – 10 m (centre to centre)
- Spacing of trusses – not more than 5 m
- Rise of the roof truss – approx 4 m
- Truss type – Pratt Truss
- Column Height – 10 m
- Yield strength of steel members = 250 MPa
- Wind speed – 55 m/s (Agartala falls in terrain category 1)
- Wind Pressure – 1.905 kN/m²

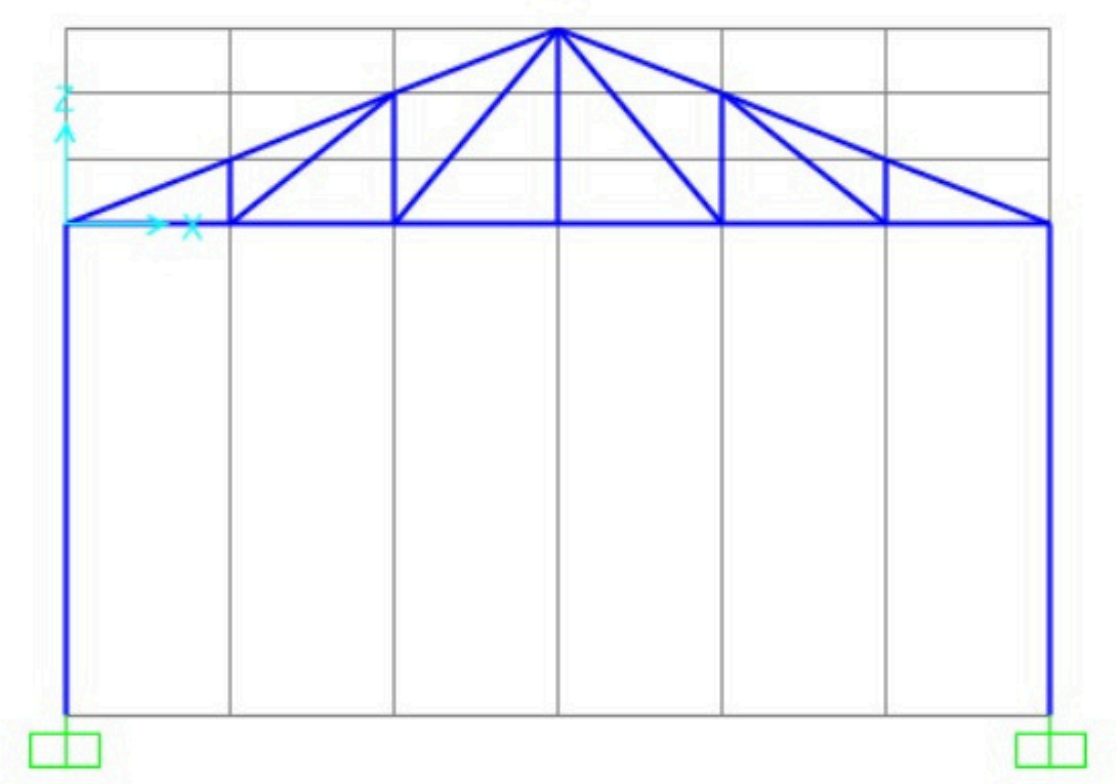
Load cases:

- Dead load – 1.5 kPa
- Live load – 1 kPa

3D View of Pratt Truss Geometry



2D View of Pratt Truss Geometry



IS: 875(Part 3): Wind Loads on Buildings and Structures – Proposed Draft & Commentary

WIND LOAD CALCULATION

5.3 – Design Wind Speed (v_z) = v_b k1 k2 k3 k4,

Where,

Basic wind speed, v_b **=55m/s**

Vz = design wind speed at any height z in m/s,

k1 = 1, probability factor (risk coefficient) (5.3.1) (Mean Probable design life of structure in years)

k2 = 1.05, terrain roughness and height factor, Terrain category 1 (5.3.2)

k3 = 1, topography factor (5.3.3)

k4 = 1.15, importance factor for the cyclonic region for industrial strs. (5.3.4).

v_z **= 1 × 1.05 × 1 × 1.15 × 55 = 66.4125 m/s**

5.4 – Design Wind Pressure

$P_z = 0.6(v_z)^2 = 0.6 \times (66.4125)^2 = 2.646$ kN/m^2

where,

P_z = wind pressure in N/m^2 at height z, and

v_z = design wind speed in m/s at height z.

The design wind pressure P_d can be obtained as,

$P_d = K_d . K_a . K_c . P_z = 1 \times 0.9 \times 0.8 \times 2.646 = 1.905$ kN/m^2

Where,

K_d = 0.9 (Wind directionality factor)

K_a =0.8 (Area averaging factor)

K_c = 1 (Combination factor)

(6.2.3.13)

Wind force (F):

$F = (C_{Pe} - C_{Pi}) \times A_e \times P_d$

h/w ≤ 0.5

$\alpha = \tan^{-}(4/10)$

$\alpha = 21.80°$

For C_{Pe} :

FOR 0°		**FOR 90°**	
EF	GH	EF	GH
- 0.33	-0.4	-0.7	-0.6

For C_{Pi} :

FOR 0°	
EF	**GH**
2	**-2**

So, (C_{Pe} – C_{Pi}) :

FOR 0°	
EF	**GH**
- 0.9	**-0.5**

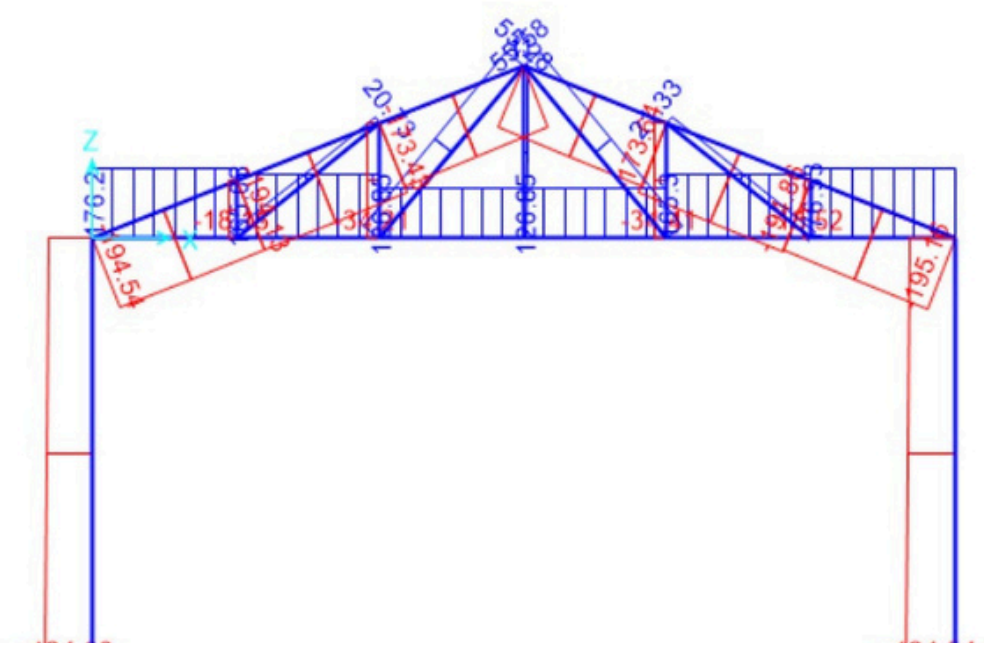
Perpendicular Distance between two purlin =

Fw = Force in the Windward direction per unit length

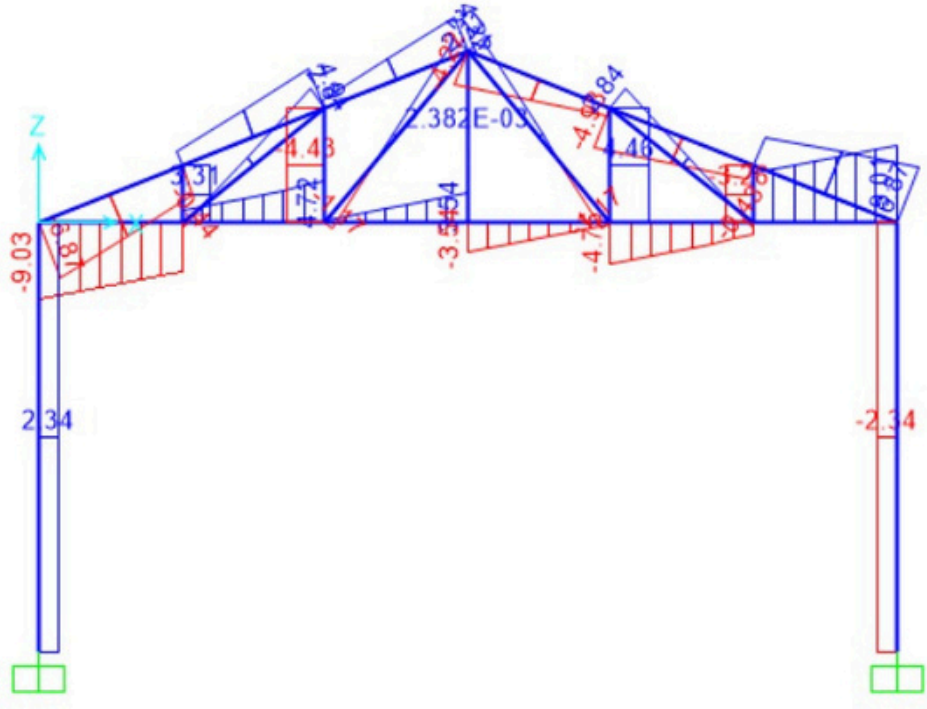
Fw = Force in the Leeward direction per unit length

Load Case = 1.5(LL+DL)

AXIAL FORCE

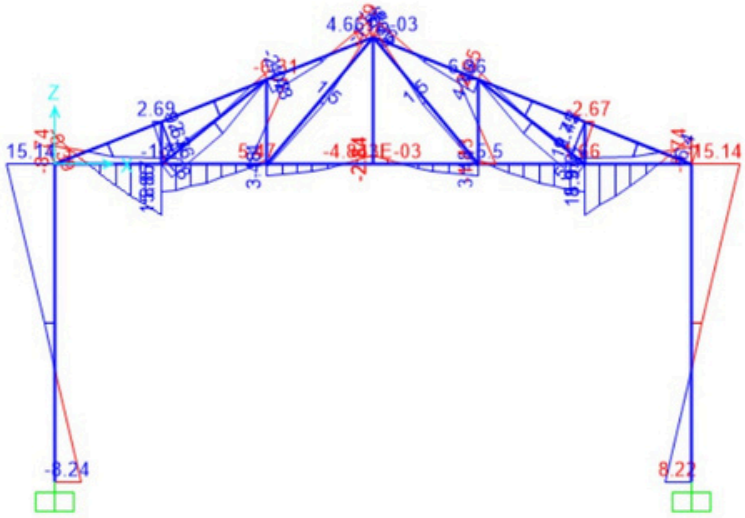


SHEAR FORCE



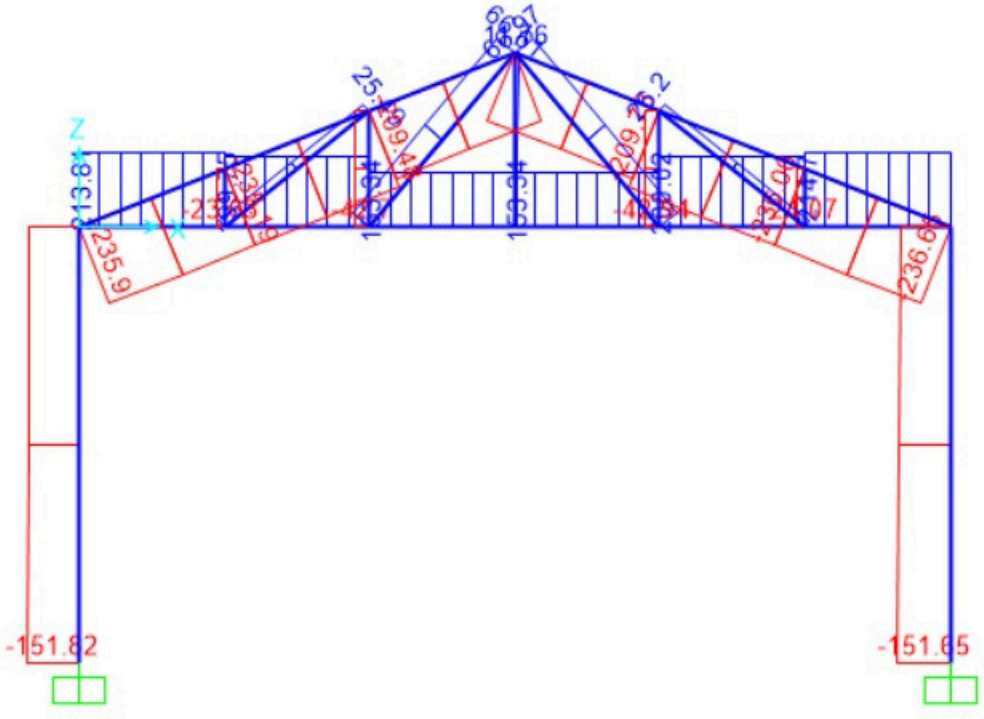


BENDING MOMENT



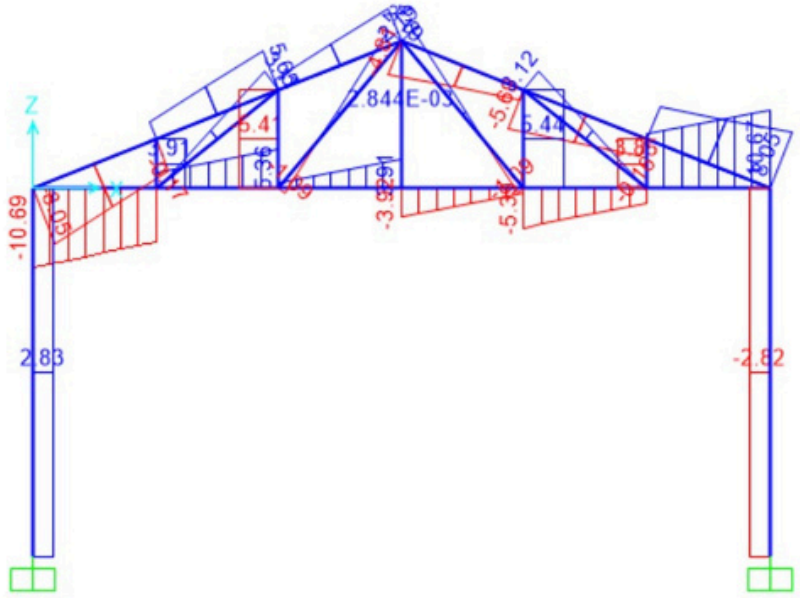
Load Case = 1.5(WL+DL)

AXIAL FORCE



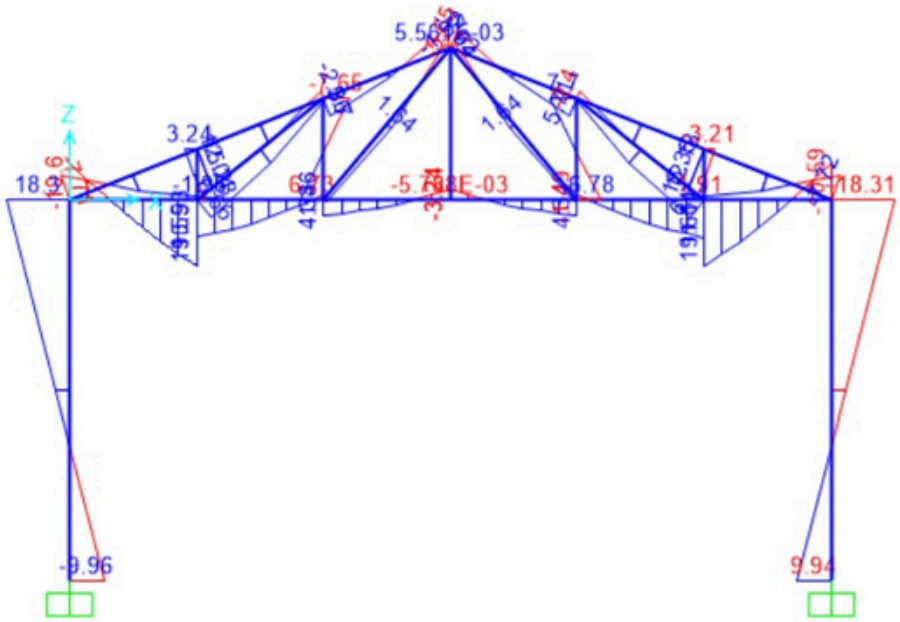
Shear Force 2-2 Diagram (1.5(DL+WL))

SHEAR FORCE

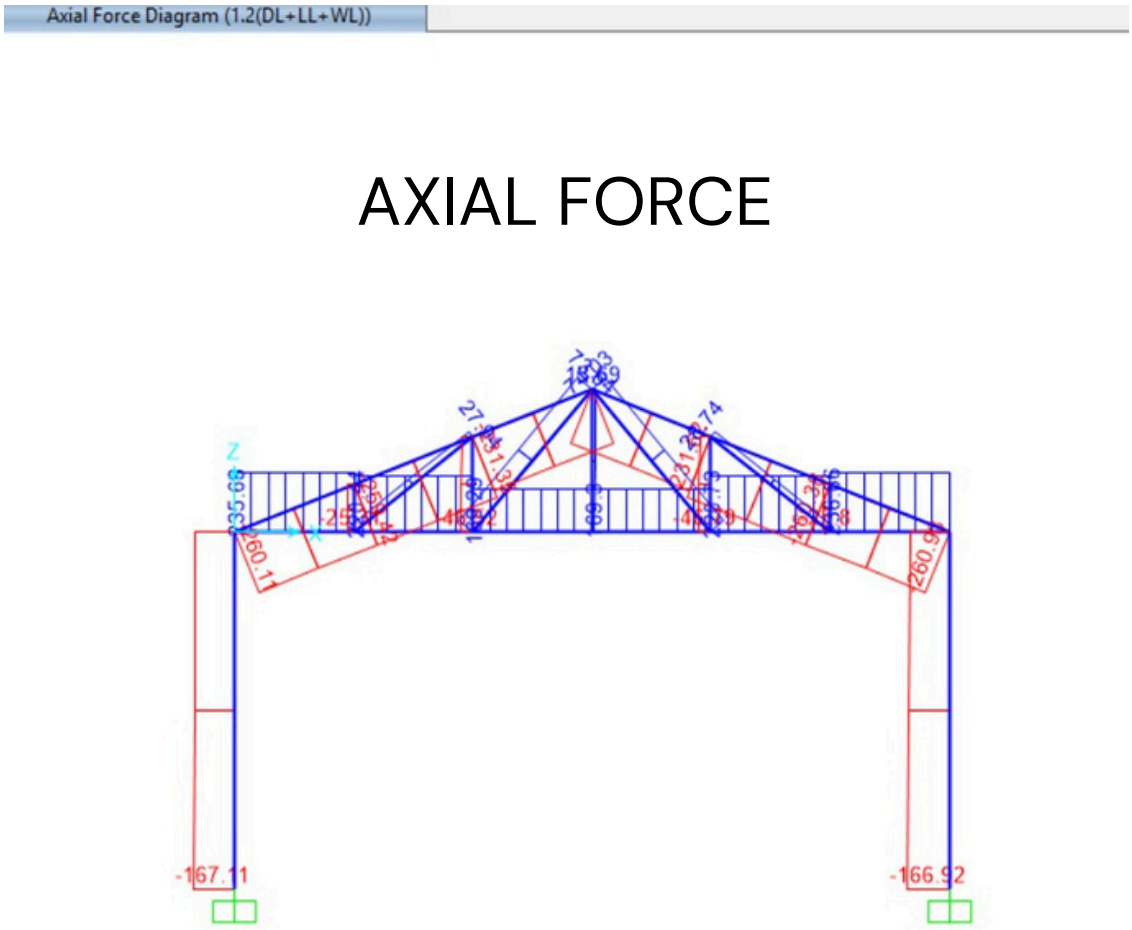


Moment 3-3 Diagram (1.5(DL+WL))

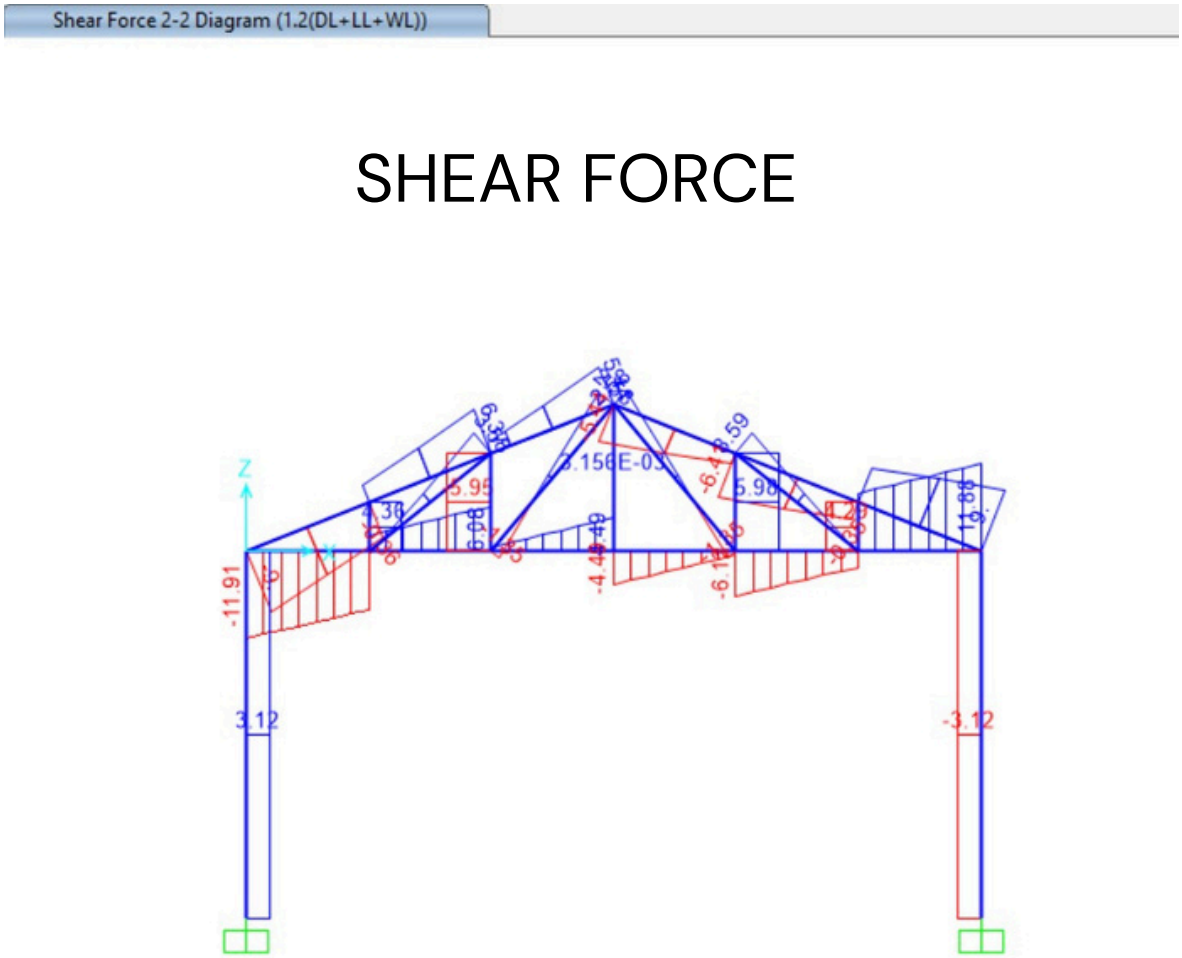
BENDING MOMENT



Load Case = 1.2(LL+DL+WL)



AXIAL FORCE



SHEAR FORCE

Moment 3-3 Diagram (1.2(DL+LL+WL))

BENDING MOMENT

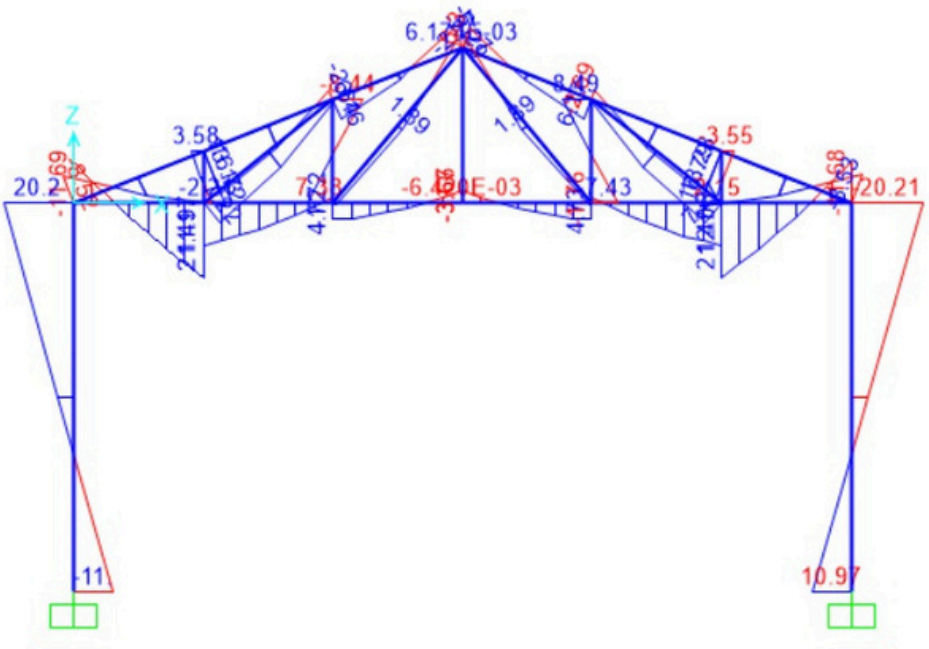


Table Forces in Critical Members

Member No.	Member	Load Type	Maximum Load (KN)	Load Case
1.	Top Chord	Axial Force	-1098.359	1.5(DL+LL)
			-1510.437	1.5(DL+WL)
			-1574.383	1.2(DL+LL+WL)
2.	Vertical	Axial Force	-222.5	1.5(DL+LL)
			-316.911	1.5(DL+WL)
			-327.306	1.2(DL+LL+WL)
3.	Diagonal	Axial Force	306.98	1.5(DL+LL)
			432.23	1.5(DL+WL)
			448.232	1.2(DL+LL+WL)
4.	Beam	Shear Force	-100.206	1.5(DL+LL)
			-130.913	1.5(DL+WL)
			-137.714	1.2(DL+LL+WL)
		Bending Moment	285.7508(kn-m)	1.5(DL+LL)
			406.2704(kn-m)	1.5(DL+WL)
			420.0579(kn-m)	1.2(DL+LL+WL)
5.	Purlin (Major)	Shear Force	-87.998	1.5(DL+LL)
			-125.545	1.5(DL+WL)
			-129.342	1.2(DL+LL+WL)
		Bending Moment	172.8284(kn-m)	1.5(DL+LL)
			-241.552(kn-m)	1.5(DL+WL)
			-250.3477(kn-m)	1.2(DL+LL+WL)
6.	Purlin (Minor)	Shear Force	-0.108	1.5(DL+LL)
			-11.545	1.5(DL+WL)
			-9.272	1.2(DL+LL+WL)
		Bending Moment	-0.3288(kn-m)	1.5(DL+LL)
			-9.946(kn-m)	1.5(DL+WL)
			-8.0663(kn-m)	1.2(DL+LL+WL)
7.	Horizontal	Axial Force	994.372	1.5(DL+LL)
			1356.182	1.5(DL+WL)
			1416.29	1.2(DL+LL+WL)
8.	Column	Axial Force	-769.066	1.5(DL+LL)
			-1040.95	1.5(DL+WL)
			-1088.91	1.2(DL+LL+WL)